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January 17, 2026

Editorial Office,
Nonlinear Dynamics,
Springer Nature

Dear Editor,

I am pleased to submit the manuscript entitled “Constructing Asymptotic Center Manifold for Extended Systems” for consideration for publication in *Nonlinear Dynamics*.

The manuscript presents a unified methodology for the approximation of center manifolds in large nonlinear systems that undergo multi-parameter perturbations and harmonic forcing through the extended systems framework, which elevates parameter perturbations and harmonic forcings to state variables. This perspective allows for the derivation of a single unified algorithm capable of treating three standard scenarios: (i) unperturbed center manifolds, (ii) parameter-perturbed systems, and (iii) harmonically forced systems.

As a key contribution the manuscript systematically explores the tangent space structures induced by the formal extension of the system, resulting in the fact that all high-order asymptotic terms of the resulting invariant manifold may be evaluated using only the resolvent of the original linearized operator. The methodology is demonstrated on a Ginzburg–Landau model at bifurcation and validated across multiple configurations of perturbation and forcing.

To the best of the author’s knowledge, this constitutes the first systematic treatment of extended systems in a general setting in the context of center manifold reduction. The author believes the manuscript will be of broad interest to the nonlinear dynamics community, particularly in areas involving high-dimensional dynamical systems, invariant manifolds, model reduction, and bifurcation theory. The author hopes that the manuscript is found suitable for consideration in *Nonlinear Dynamics*.

Sincerely,

Best Regards

Prabal S. Negi